QUOT OF O^{OPLUS R} ON P^N_Z WITH HILBERT POLYNOMIAL P IS PROPER OVER Z

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1046-1056 of sources/stacks-moduli/pinned-original-source.tex; proof span is lines 1058-1190, the complete author proof (131 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 1 of queue_moduli.json (pb=4649, stmt 1046-1056, proof 1058-1190).

Editorial note. stacks-moduli wave-1: fresh adjudication accept (2026-10-14); pool per Chief order 08:5x (dpa superseded to lane-s9); moduli.tex 53 labels/48 with proof judged 25 accept / 23 reject / 5 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0DM4: sources/stacks-moduli/pinned-original-source.tex, source lines 770–776. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DM3: sources/stacks-moduli/pinned-original-source.tex, source lines 755–759. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0ASY: sources/stacks-moduli/pinned-original-source.tex, source lines 1190–1190. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0A40: sources/stacks-moduli/pinned-original-source.tex, source lines 1190–1190. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08AG: sources/stacks-moduli/pinned-original-source.tex, source lines 1190–1190. Cited by the retained statement or proof but

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not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08A8: sources/stacks-moduli/pinned-original-source.tex, source lines 1190–1190. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08A6: sources/stacks-moduli/pinned-original-source.tex, source lines 1190–1190. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $n \geq 0$, $r \geq 1$, $P \in \mathbf{Q}[t]$. The algebraic space*

$$X = \mathrm{Quot}^P_{\mathcal{O}_{\mathbf{P}^n_{\mathbf{Z}}}^{\oplus r}/\mathbf{P}^n_{\mathbf{Z}}/\mathbf{Z}}$$

parametrizing quotients of $\mathcal{O}_{\mathbf{P}^n_{\mathbf{Z}}}^{\oplus r}$ with Hilbert polynomial P is proper over $\mathrm{Spec}(\mathbf{Z})$.

Proof. We already know that $X \to \mathrm{Spec}(\mathbf{Z})$ is separated and locally of finite presentation (Lemma 0DM3). We also know that $X \to \mathrm{Spec}(\mathbf{Z})$ satisfies the existence part of the valuative criterion, see Lemma 0DM4. By the valuative criterion for properness, it suffices to prove our Quot space is quasi-compact, see Morphisms of Spaces, Lemma 0A40. Thus it suffices to find a quasi-compact scheme T and a surjective morphism $T \to X$. Let m be the integer found in Varieties, Lemma 08AG. Let

$$N = r\binom{m+n}{n} - P(m)$$

We will write $\mathbf{P}^n$ for $\mathbf{P}^n_{\mathbf{Z}} = \mathrm{Proj}(\mathbf{Z}[T_0, \ldots, T_n])$ and unadorned products will mean products over $\mathrm{Spec}(\mathbf{Z})$. The idea of the proof is to construct a "universal" map

$$\Psi : \mathcal{O}_{T\times\mathbf{P}^n}(-m)^{\oplus N} \longrightarrow \mathcal{O}_{T\times\mathbf{P}^n}^{\oplus r}$$

over an affine scheme T and show that every point of X corresponds to a cokernel of this in some point of T.

Definition of T and Ψ. We take $T = \mathrm{Spec}(A)$ where

$$A = \mathbf{Z}[a_{i,j,E}]$$

where $i \in \{1, \ldots, r\}$, $j \in \{1, \ldots, N\}$ and $E = (e_0, \ldots, e_n)$ runs through the multi-indices of total degree $|E| = \sum_{k=0,\ldots n} e_k = m$. Then we define Ψ to be the map whose (i, j) matrix entry is the map

$$\sum\nolimits_{E=(e_0,\ldots,e_n)} a_{i,j,E} T_0^{e_0} \ldots T_n^{e_n} : \mathcal{O}_{T\times\mathbf{P}^n}(-m) \longrightarrow \mathcal{O}_{T\times\mathbf{P}^n}$$

where the sum is over E as above (but i and j are fixed of course).

Consider the quotient $\mathcal{Q} = \mathrm{Coker}(\Psi)$ on $T \times \mathbf{P}^n$. By More on Morphisms, Lemma 0ASY there exists a $t \geq 0$ and closed subschemes

$$T = T_0 \supset T_1 \supset \ldots \supset T_t = \emptyset$$

such that the pullback $\mathcal{Q}_p$ of $\mathcal{Q}$ to $(T_p \setminus T_{p+1}) \times \mathbf{P}^n$ is flat over $T_p \setminus T_{p+1}$. Observe that we have an exact sequence

$$\mathcal{O}_{(T_p\setminus T_{p+1})\times\mathbf{P}^n}(-m)^{\oplus N} \to \mathcal{O}_{(T_p\setminus T_{p+1})\times\mathbf{P}^n}^{\oplus r} \to \mathcal{Q}_p \to 0$$

by pulling back the exact sequence defining $\mathcal{Q} = \mathrm{Coker}(\Psi)$. Therefore we obtain a morphism

$$\coprod (T_p \setminus T_{p+1}) \longrightarrow \mathrm{Quot}_{\mathcal{O}^{\oplus r}/\mathbf{P}/\mathbf{Z}} \supset \mathrm{Quot}^P_{\mathcal{O}^{\oplus r}/\mathbf{P}/\mathbf{Z}} = X$$

Since the left hand side is a Noetherian scheme and the inclusion on the right hand side is open, it suffices to show that any point of X is in the image of this morphism.

Let k be a field and let $x \in X(k)$. Then x corresponds to a surjection $\mathcal{O}_{\mathbf{P}^n_k}^{\oplus r} \to \mathcal{F}$ of coherent $\mathcal{O}_{\mathbf{P}^n_k}$-modules such that the Hilbert polynomial of $\mathcal{F}$ is P. Consider the short exact sequence

$$0 \to \mathcal{K} \to \mathcal{O}_{\mathbf{P}^n_k}^{\oplus r} \to \mathcal{F} \to 0$$

By Varieties, Lemma 08AG and our choice of m we see that $\mathcal{K}$ is m-regular. By Varieties, Lemma 08A8 we see that $\mathcal{K}(m)$ is globally generated. By Varieties, Lemma 08A6 and the definition of m-regularity we see that $H^i(\mathbf{P}^n_k, \mathcal{K}(m)) = 0$ for $i > 0$. Hence we see that

$$\dim_k H^0(\mathbf{P}^n_k, \mathcal{K}(m)) = \chi(\mathcal{K}(m)) = \chi(\mathcal{O}_{\mathbf{P}^n_k}(m)^{\oplus r}) - \chi(\mathcal{F}(m)) = N$$

by our choice of N. This gives a surjection

$$\mathcal{O}_{\mathbf{P}^n_k}^{\oplus N} \longrightarrow \mathcal{K}(m)$$

Twisting back down and using the short exact sequence above we see that $\mathcal{F}$ is the cokernel of a map

$$\Psi_x : \mathcal{O}_{\mathbf{P}^n_k}(-m)^{\oplus N} \to \mathcal{O}_{\mathbf{P}^n_k}^{\oplus r}$$

There is a unique ring map $\tau : A \to k$ such that the base change of Ψ by the corresponding morphism $t = \mathrm{Spec}(\tau) : \mathrm{Spec}(k) \to T$ is Ψ_x. This is true because the entries of the $N \times r$ matrix defining Ψ_x are homogeneous polynomials $\sum \lambda_{i,j,E} T_0^{e_0} \ldots T_n^{e_n}$ of degree m in $T_0, \ldots, T_n$ with coefficients $\lambda_{i,j,E} \in k$ and we can set $\tau(a_{i,j,E}) = \lambda_{i,j,E}$. Then $t \in T_p \setminus T_{p+1}$ for some p and the image of t under the morphism above is x as desired. $\square$